

Lecture 3

More Simple Linear Regression

DSC 40A, Summer 2024

Announcements

- Homework 1 is due **tonight**, 11:59PM PT.
- Homework 2 is due tomorrow, **Tuesday**, at 11:59PM PT.
 - We are aiming to give you feedback on HW 1 before HW 2 is due on Tuesday, keep an eye out tomorrow.
- Week 2 picks up the pace really quickly, please prepare to have more of a workload this week as we start prepare for your upcoming exam.
- Groupwork 1 solutions are available on [Piazza](#).

(Community) OH TODAY!

2:00 pm PT
here

Exam Announcements

- Midterm Exam Logistics will be released this week, please keep an eye on your inbox for those. We will be doing live proctoring and you will be expected to be on Zoom with camera and sound on. More details to come ASAP.
 - Many of you are in non-PT time zones. Please plan to take the exam during the expected exam time slot: Wednesday, July 15th, 11:00AM-1:50PM, PT.
 - We will **not** be accommodating different exam times, unless you have OSD accommodations specifically for that.
- If you intend to use your exam accommodations on this upcoming midterm, please make sure OSD sends your AFA letter to us **by this Friday, July 10th**.
 - It's hard to change/reschedule your exam last minute, so giving us at least the weekend to rearrange and make sure we can accommodate you is really helpful.

Agenda - Part 1

- Recap: Simple linear regression.
- Correlation.
- Interpreting the formulas.
- Connections to related models.
- Introduction to linear algebra

← spend more time on interpreting



Lecture 3 Part 1

More Simple Linear Regression

DSC 40A, Summer 2026

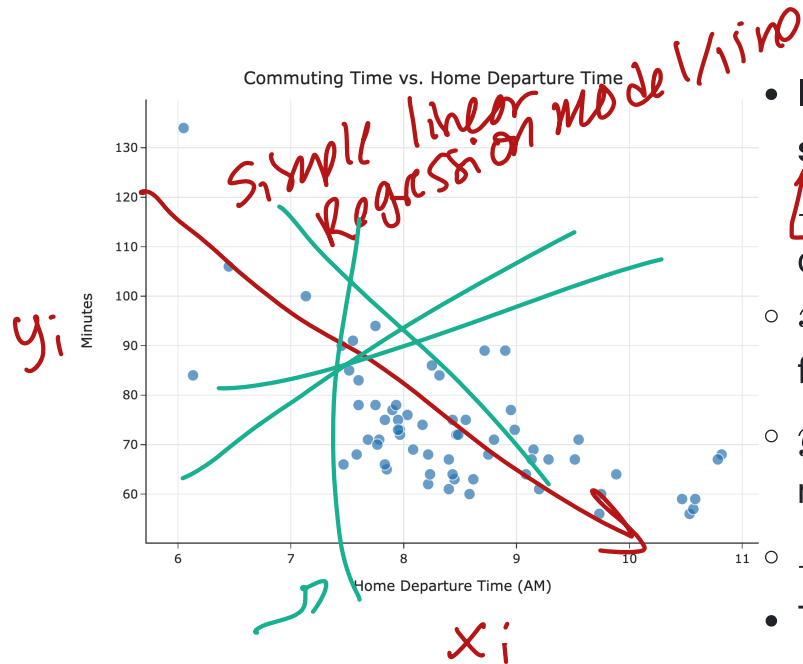
Question 🤔

Ask in Q&A, respond to questions in chat.

Remember, you can always ask questions!

Recap: Simple linear regression

Recap



$$(y_i - \hat{y}_i)^2$$

- In Lecture 2 part 1 our goal was to fit a simple linear regression model, $H(x) = w_0 + w_1 x$, to our commute times dataset. *any lines*
- x_i : The i th home departure time (e.g. 8.5, for 8:30 AM).
- y_i : The i th actual commute time (e.g. 76 minutes).
- $H(x_i)$: The i th predicted commute time.
- To do so, we used squared loss.

which is the best?

calculus

Least squares solutions

- Our goal was to find the parameters w_0^* and w_1^* that minimized:

partial
derivatives

$$R_{sq}(w_0, w_1) = \frac{1}{n} \sum_{i=1}^n (y_i - (w_0 + w_1 x_i))^2$$

$$y = w_1 x + w_0$$

- To do so, we used calculus, and we found that the minimizing values are:

Best slope

$$w_1^* = \frac{\sum_{i=1}^n (x_i - \bar{x})(y_i - \bar{y})}{\sum_{i=1}^n (x_i - \bar{x})^2}$$

Best intercept

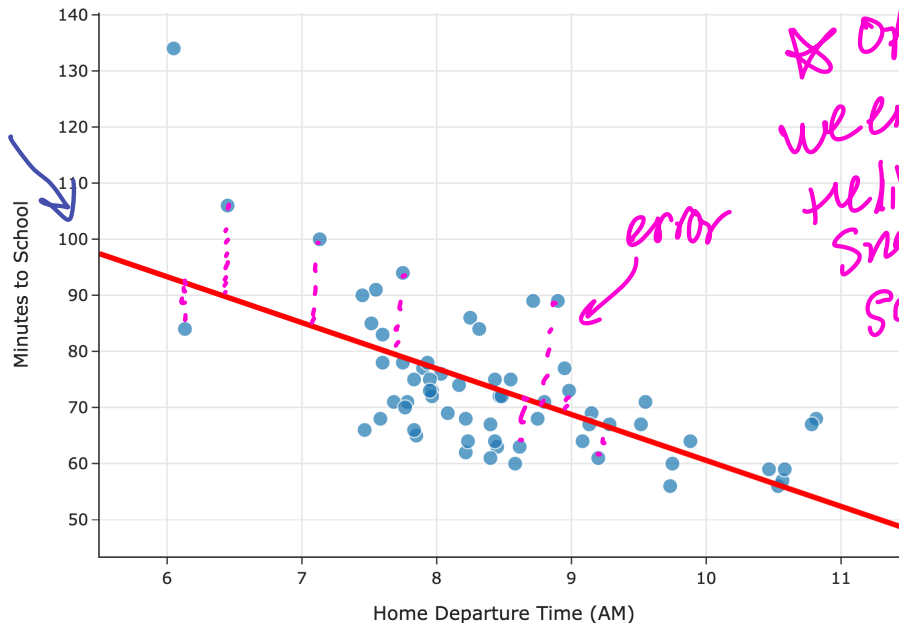
$$w_0^* = \bar{y} - w_1^* \bar{x}$$

Best slope

- We say w_0^* and w_1^* are **optimal parameters**, and the resulting line is called the **regression line**.

w_0^* optimal intercept w_1^* optimal slope

Predicted Commute Time = $142.25 - 8.19 * \text{Departure Hour}$



* optimal parameters
we end up with
the line with the
smallest average
squared error

↑
goal to find

LS

Now what?

We've found the optimal slope and intercept for linear hypothesis functions using squared loss (i.e. for the regression line). Now, we'll:

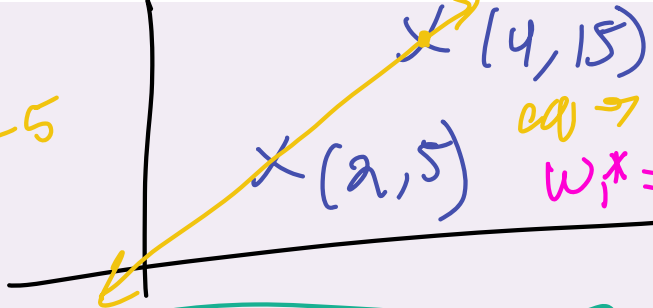
correlation
coefficient

- See how the formulas we just derived connect to the formulas for the slope and intercept of the regression line we saw in DSC 10.
 - They're the same, but we need to do a bit of work to prove that.
- Learn how to interpret the slope of the regression line.
- Understand connections to other related models.
- Learn how to build regression models with **multiple inputs**.
 - To do this, we'll need linear algebra!

intercept
 $4(5) + w_0 = 15$
 $\times$ slope $w_0 = -5$

Question 🤔

Answer in chat



easy $\Rightarrow$ smallest MSE
 $w_1^* = \text{slope} = \frac{y_2 - y_1}{x_2 - x_1} = \frac{15 - 5}{4 - 2} = \frac{10}{2} = 5$
 $H(x) = w_0 + w_1 x$
 regression line

Consider a dataset with just two points, (2, 5) and (4, 15). Suppose we want to fit a linear hypothesis function to this dataset using squared loss. What are the values of w_0^* and w_1^* that minimize empirical risk?

- A. $w_0^* = 2, w_1^* = 5$
- ~~B. $w_0^* = 3, w_1^* = 10$~~
- C. $w_0^* = -2, w_1^* = 5$
- D. $w_0^* = -5, w_1^* = 5$

easy
 given any 2 pts $\Rightarrow$ can always
 find a line that goes thru
 those 2 pts.
 MSE = 0 if goes directly
 thru the pts

Hard
 use formulas
 if only 2 pts

Correlation

Association: only pattern

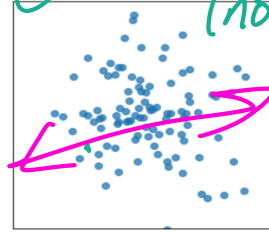
correlation: linear association pattern that looks like a line

Quantifying patterns in scatter plots

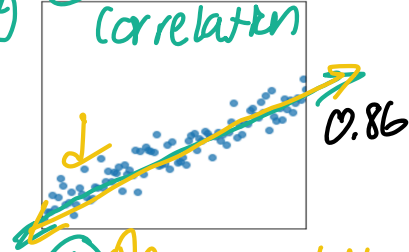
- In DSC 10, you were introduced to the idea of the **correlation coefficient**, r .
- It is a measure of the strength of the **linear association** of two variables, x and y . *↓ how much did it look like a line*
- Intuitively, it measures how tightly clustered a scatter plot is around a straight line.
- It ranges between -1 and 1.

$r \rightarrow$ negative: negative linear association
 $\rightarrow$ positive: positive linear association

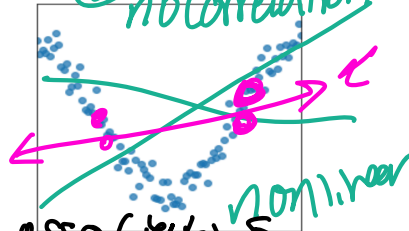
① No association (no correlation)



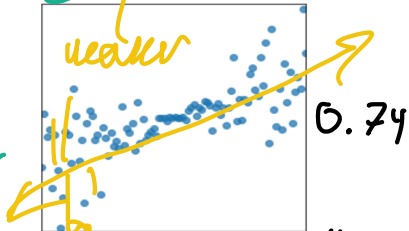
② Strong pos correlation



③ association no correlation



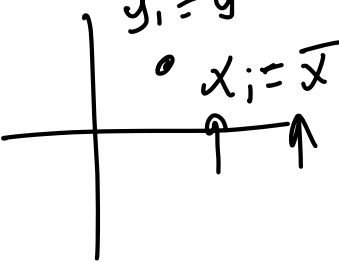
④ pos correlation weaker



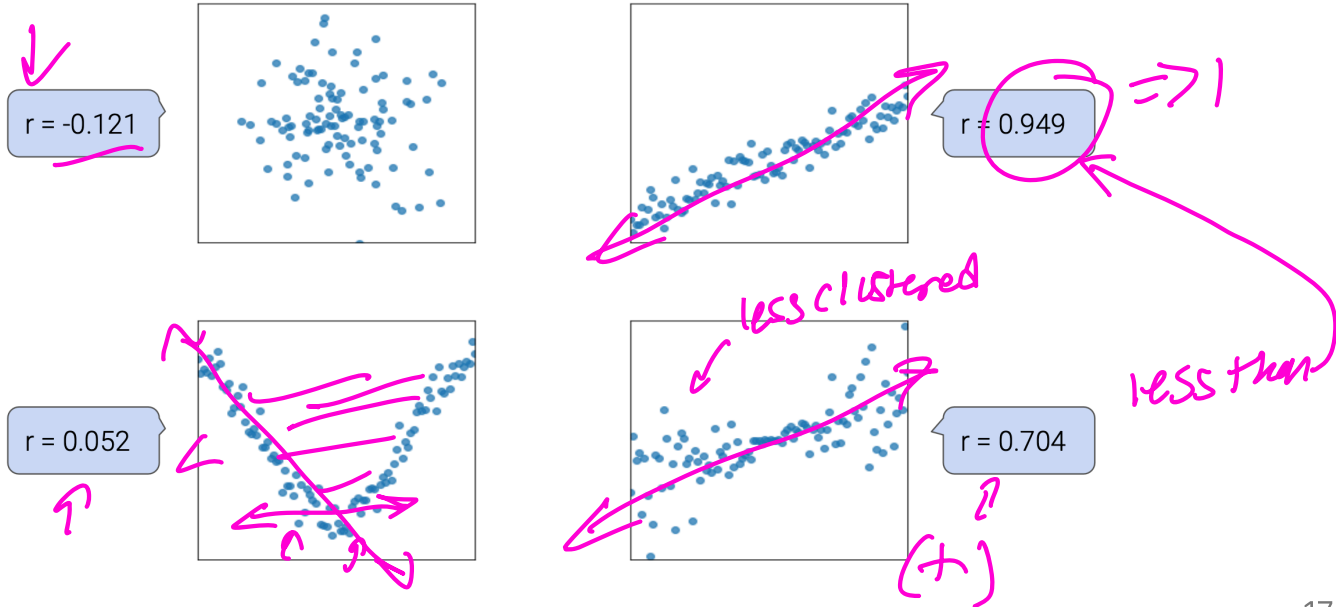
the closer to 1 the stronger the correlation

pearson's correlation coefficient (here ~~correlation~~)

The correlation coefficient

- code
DSC
10
- The correlation coefficient (r) is defined as the average of the product of x and y , when both are in standard units.
 - Let σ_x be the standard deviation of the x_i s, and $\bar{x}$ be the mean of the x_i s.
 - x_i in standard units is $\frac{x_i - \bar{x}}{\sigma_x}$.
Value - mean $\rightarrow$ measure # of SDs above or below the mean
SD
 - The correlation coefficient, then, is:
- yi = y*
x_i = x̄
- 
- $$r = \frac{1}{n} \sum_{i=1}^n \left(\frac{x_i - \bar{x}}{\sigma_x} \right) \left(\frac{y_i - \bar{y}}{\sigma_y} \right)$$
- average* x_i in SU y_i in SU
- yi = ȳ or xi = x̄*
0 + 0 + 0

The correlation coefficient, visualized



Another way to express w_1^*

- It turns out that w_1^* , the optimal slope for the linear hypothesis function when using squared loss (i.e. the regression line), can be written in terms of r !

Slope

$$w_1^* = \frac{\sum_{i=1}^n (x_i - \bar{x})(y_i - \bar{y})}{\sum_{i=1}^n (x_i - \bar{x})^2} r \frac{\sigma_y}{\sigma_x}$$

correlation coeff
to find the optimal
model parameter

- It's not surprising that r is related to w_1^* , since r is a measure of linear association.
- Concise way of writing w_0^* and w_1^* :

$$w_1^* = r \frac{\sigma_y}{\sigma_x} \quad w_0^* = \bar{y} - w_1^* \bar{x}$$

Proof that

$$w_1^* =$$

$$r \frac{\sigma_y}{\sigma_x}$$

Goal

w/ sum w/ ring

$$\textcircled{1} w_1^* = \frac{\sum_{i=1}^n (x_i - \bar{x})(y_i - \bar{y})}{\sum_{i=1}^n (x_i - \bar{x})^2}$$

σ_x

$$= \frac{r n \sigma_x \sigma_y}{n \sigma_x^2}$$

$\textcircled{6}$ cancel junk out out

$$= \frac{r \sigma_y}{\sigma_x}$$

Goal?

$\textcircled{2}$ Aside

"product of x & y in sum"

$$r = \frac{1}{n} \sum_{i=1}^n \left(\frac{x_i - \bar{x}}{\sigma_x} \right) \left(\frac{y_i - \bar{y}}{\sigma_y} \right)$$

→ rearrange this formula

$\textcircled{3}$ simplify?

SD of x & y (σ_x, σ_y)

$$r = \frac{1}{n} \cdot \frac{1}{\sigma_x \sigma_y} \sum_{i=1}^n (x_i - \bar{x})(y_i - \bar{y})$$

$$\textcircled{A} \sum_{i=1}^n (x_i - \bar{x})(y_i - \bar{y}) = r n \sigma_x \sigma_y$$

Proof that $w_1^* = r \frac{\sigma_y}{\sigma_x}$

④ write denominator
to cancel out junk we don't
want in numerator

variance

⑤

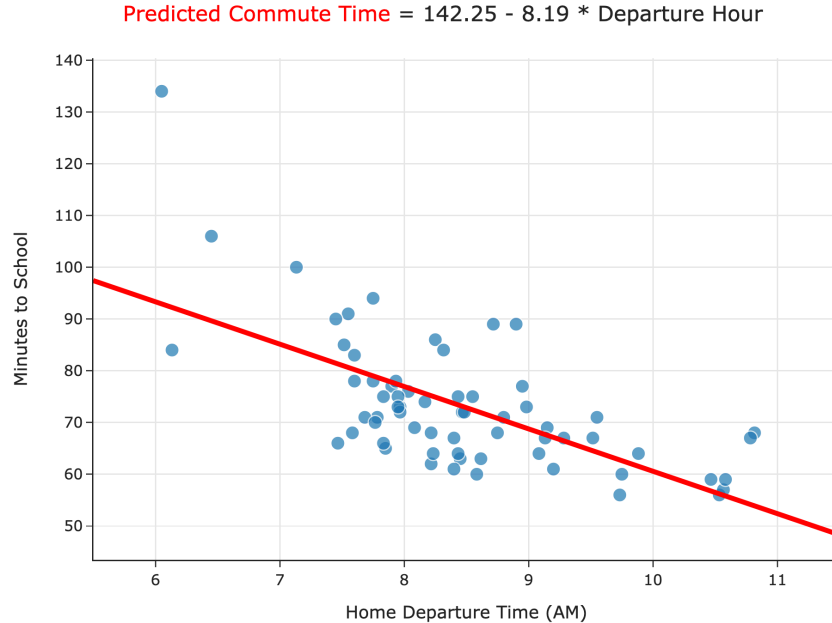
$$\sigma_x = \sqrt{\frac{1}{n} \sum_{i=1}^n (x_i - \bar{x})^2}$$

$$\text{denom} = \sum_{i=1}^n (x_i - \bar{x})^2$$

$$\sigma_x^2 = \frac{1}{n} \sum_{i=1}^n (x_i - \bar{x})^2$$

$$\underline{n \sigma_x^2} = \sum_{i=1}^n (x_i - \bar{x})^2$$

Let's test these new formulas out in code! Follow along [here](#).



Back @ 12:30

Interpreting the formulas

Interpreting the slope

no units $\swarrow$ $\searrow$ units of y

$$w_1^* = r \frac{\sigma_y}{\sigma_x}$$

$\searrow$ units of x

- The units of the slope are **units of y per units of x** .
- In our commute times example, in $H(x) = 142.25 - 8.19x$, our predicted commute time **decreases by 8.19 minutes per hour**.

x_i : departure time in hrs

y_i : commute times in mins

8 mins 12 sec

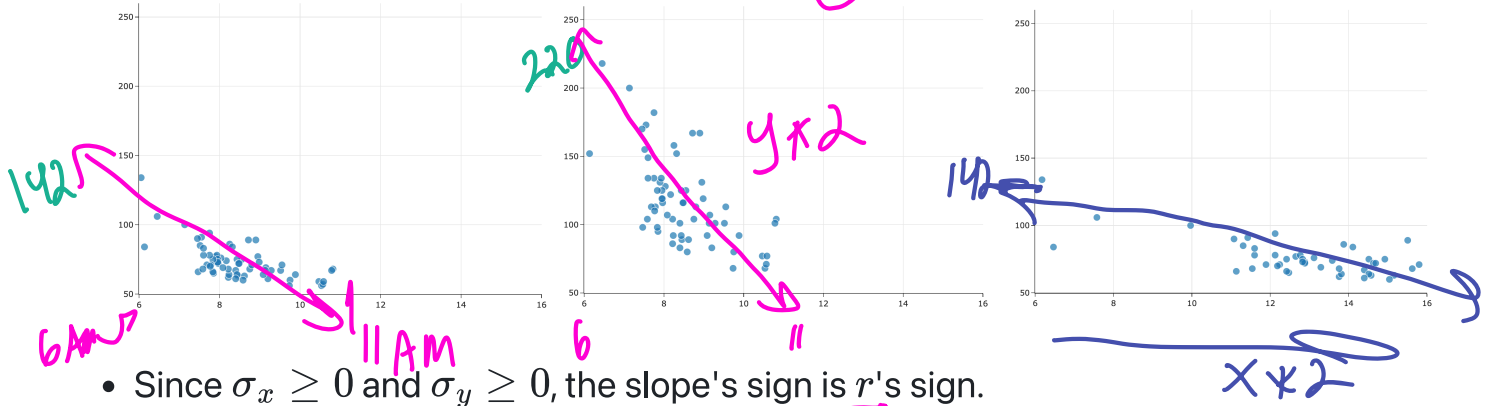
$$(.20 \cdot 60 \approx 12)$$

Interpreting the slope

$$w_1^* = r \frac{\sigma_y}{\sigma_x} \geq 0$$

$\swarrow$ slope $\uparrow$
 $\searrow$ $\sigma_y \geq 0$
 $\searrow$ $\sigma_x \geq 0$

bigger fraction gets bigger
 bigger, fraction gets smaller
 so slope is shallower

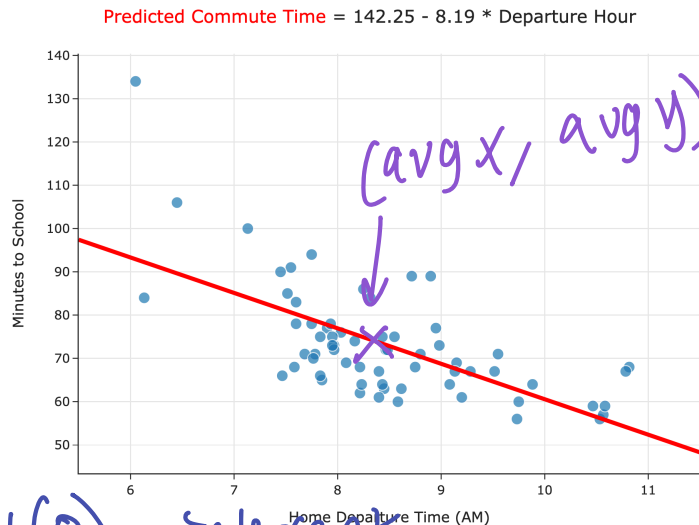


• Since $\sigma_x \geq 0$ and $\sigma_y \geq 0$, the slope's sign is r 's sign.

✓ • As the y values get more spread out, σ_y increases, so the slope gets steeper.

✗ • As the x values get more spread out, σ_x increases, so the slope gets shallower.

Interpreting the intercept



units of y - units of x • units of x

$w_0^* = \bar{y} - w_1^* \bar{x}$

- What are the units of the intercept?

whatever units of y or x
mins

- What is the value of $H^*(\bar{x})$?

$$H^*(x_i) = w_0^* + w_1^* x_i$$

$$= \bar{y} - w_1^* \bar{x} + w_1^* x_i$$

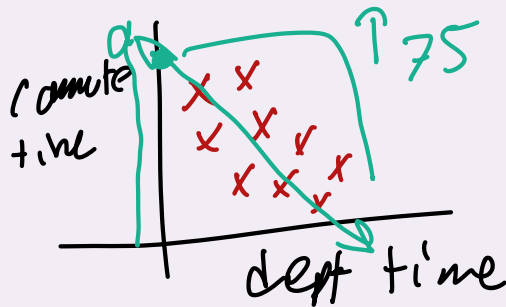
$$H^*(\bar{x}) = \bar{y} + w_1^* (\bar{x} - \bar{x})$$

$$H^*(\bar{x}) = \bar{y}$$

$H(0) = \text{intercept}$
= predicted commute time at midnight

Question 🤔

Answer in chat.



We fit a regression line to predict commute times given departure hour. Then, we add 75 minutes to all commute times in our dataset. What happens to the resulting regression line?

- • A. Slope increases, intercept increases.
- B. Slope decreases, intercept increases.
- • C. Slope stays the same, intercept increases.
- D. Slope stays the same, intercept stays the same.

depends on r, σ_x, σ_y
corr coeff: how much
does it look like a line

→ how to prove?

Correlation and mean squared error (Proof in FAQs page)

- **Claim:** Suppose that w_0^* and w_1^* are the optimal intercept and slope for the regression line. Then,

$$\text{MSE } R_{\text{sq}}(w_0^*, w_1^*) = \sigma_y^2(1 - r^2)$$

- That is, the mean squared error of the regression line's predictions and the correlation coefficient, r , always satisfy the relationship above.
- Even if it's true, why do we care?
 - In machine learning, we often use both the mean squared error and r^2 to compare the performances of different models.
 - If we can prove the above statement, we can show that **finding models that minimize mean squared error** is equivalent to **finding models that maximize r^2**

Connections to related models

Question 🤔

Answer in chat.

Suppose we chose the model $H(x) = w_1 x$ and squared loss.

What is the optimal model parameter, w_1^* ? $\hookrightarrow w_0? = 0$

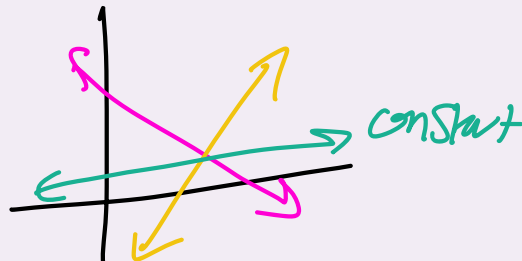
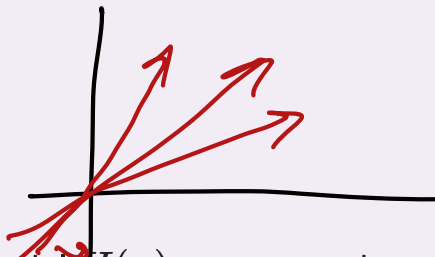
• A.
$$\frac{\sum_{i=1}^n (x_i - \bar{x})(y_i - \bar{y})}{\sum_{i=1}^n (x_i - \bar{x})^2}$$

• B.
$$\frac{\sum_{i=1}^n x_i y_i}{\sum_{i=1}^n x_i^2}$$

• C.
$$\frac{\sum_{i=1}^n x_i y_i}{\sum_{i=1}^n x_i^2}$$

• D.
$$\frac{\sum_{i=1}^n y_i}{\sum_{i=1}^n x_i}$$

Typo
Same thing

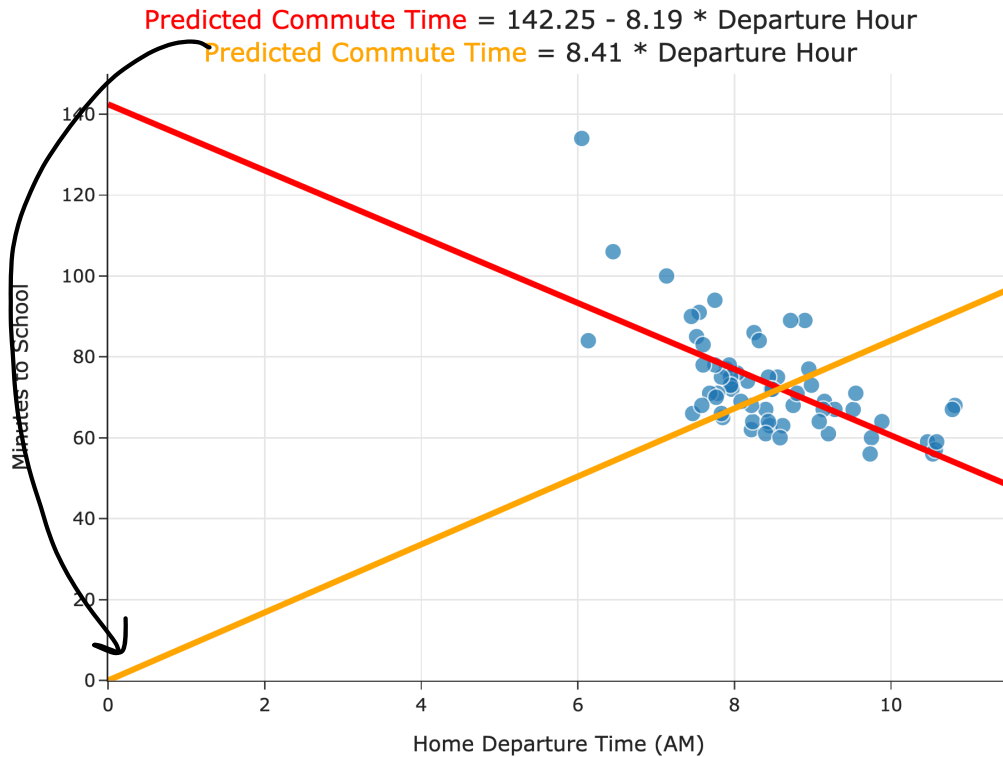


Exercise

Suppose we chose the model $H(x) = w_1 x$ and squared loss.

What is the optimal model parameter, w_1^* ? $\rightarrow$ calculus + derivation

$$R_{sq}(w_1) = \frac{1}{n} \sum (y_i - w_1 x_i)^2$$



Exercise

Suppose we choose the model $H(x) = w_0$ and squared loss.

What is the optimal model parameter, w_0^* ?

re-solve?
No!

constant model

$$h = w_0$$

$$w_0^* = \text{mean}(x_1, x_2, \dots, x_n)$$

Comparing mean squared errors

- With both:
 - the constant model, $H(x) = h$, and
 - the simple linear regression model, $H(x) = w_0 + w_1x$,

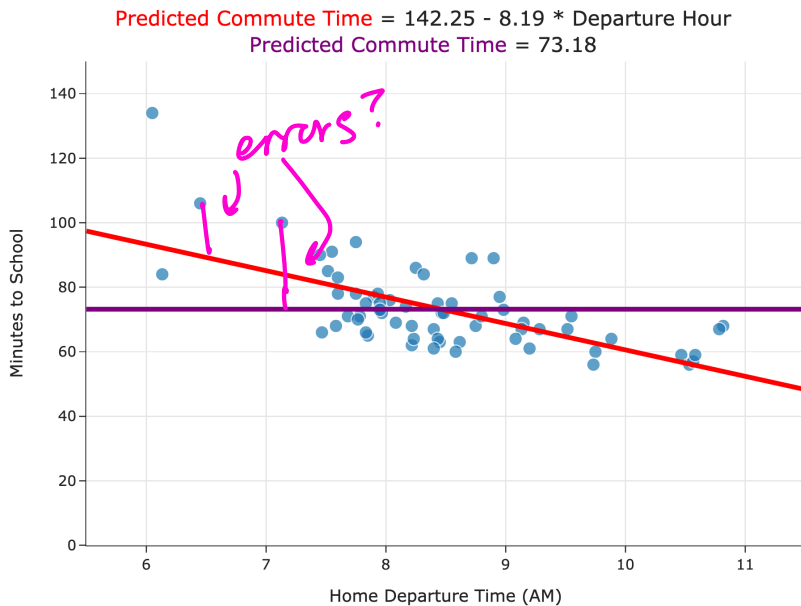
when we chose squared loss, we minimized mean squared error to find optimal parameters:

$$\text{MSE} \\ R_{\text{sq}}(H) = \frac{1}{n} \sum_{i=1}^n (y_i - H(x_i))^2$$

- Which model minimizes mean squared error more?

Linear regression $\rightarrow$ can become the constant model
but constant can't switch... (slope=0)

Comparing mean squared errors



$$\text{MSE} = \frac{1}{n} \sum_{i=1}^n (y_i - H(x_i))^2$$

- The MSE of the best simple linear regression model is ≈ 97 .
- The MSE of the best constant model is ≈ 167 . *variance*
- The simple linear regression model is a more flexible version of the constant model.

$$\text{MSE}_{\text{SLRM}} < \text{MSE}_{\text{constant}}$$

Linear algebra review

Wait... why do we need linear algebra?

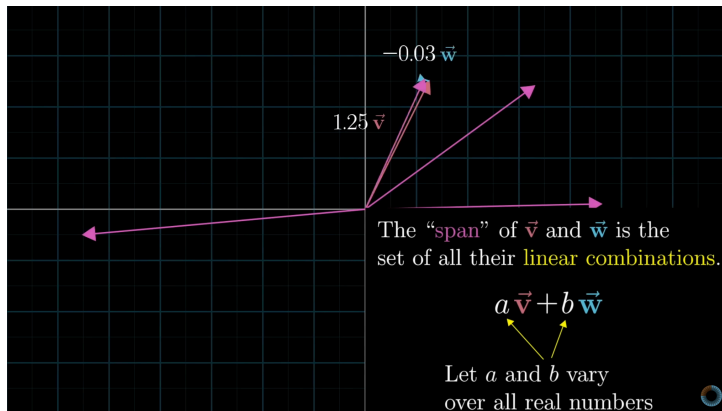
- Soon, we'll want to make predictions using more than one feature.
 - Example: Predicting commute times using departure hour and temperature.
- Thinking about linear regression in terms of **matrices and vectors** will allow us to find hypothesis functions that:
 - Use multiple features (input variables).
 - Are non-linear, e.g. $H(x) = w_0 + w_1x + w_2x^2$.
- Before we dive in, let's review.

Spans of vectors

- One of the most important ideas you'll need to remember from linear algebra is the concept of the **span** of two or more vectors.
- To jump start our review of linear algebra, let's start by watching  [this video by](#)

~~3blue1brown.~~

Linear
algebra



Next time

- We'll review the necessary linear algebra prerequisites.
- We'll then start to formulate the problem of minimizing mean squared error for the simple linear regression model **using matrices and vectors**.

Break

Lecture 3 Part 2

Dot Products and Projections

DSC 40A, Summer 2026

Agenda for Part 2

- Recap: Friends of simple linear regression.
- Dot products.
- Spans and projections.

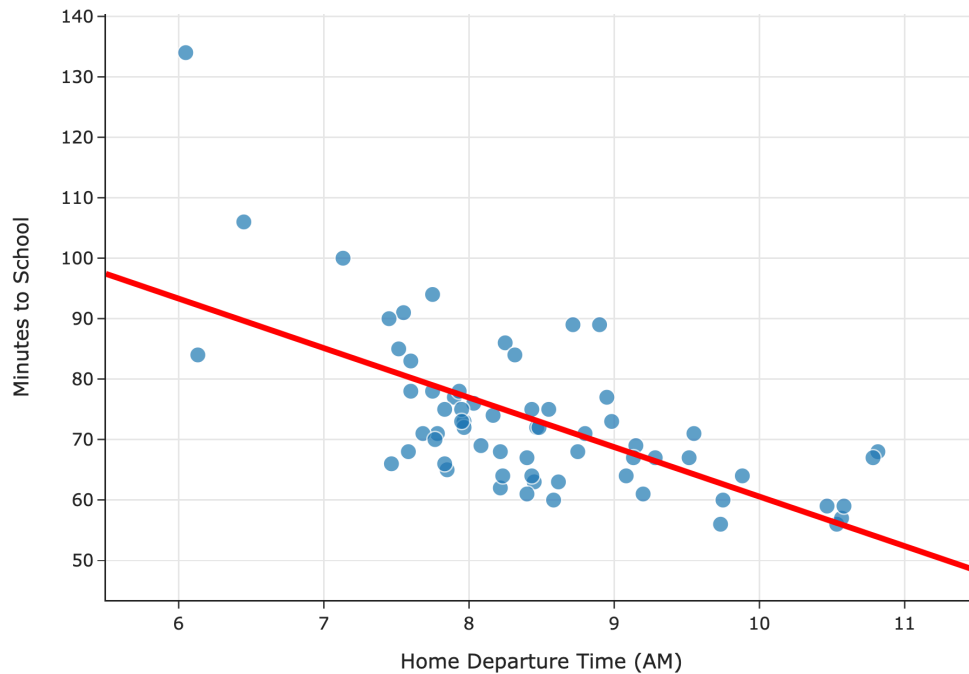
Question 🤔

Answer in chat

Remember, you can always ask questions!

Recap: Friends of simple linear regression

Predicted Commute Time = $142.25 - 8.19 * \text{Departure Hour}$



Simple linear regression

- Model: $H(x) = w_0 + w_1x$.
- Loss function: squared loss, i.e. $L_{\text{sq}}(y_i, H(x_i)) = (y_i - H(x_i))^2$.
- Average loss, i.e. empirical risk:

$$R_{\text{sq}}(w_0, w_1) = \frac{1}{n} \sum_{i=1}^n (y_i - (w_0 + w_1x_i))^2$$

- Optimal model parameters, found by minimizing empirical risk:

$$w_1^* = \frac{\sum_{i=1}^n (x_i - \bar{x})(y_i - \bar{y})}{\sum_{i=1}^n (x_i - \bar{x})^2} = r \frac{\sigma_y}{\sigma_x} \qquad w_0^* = \bar{y} - w_1^* \bar{x}$$

Friends of simple linear regression

- Suppose we use squared loss throughout.
- If our model is $H(x) = w_1x$, it is a **line that is forced through the origin, $(0, 0)$.**

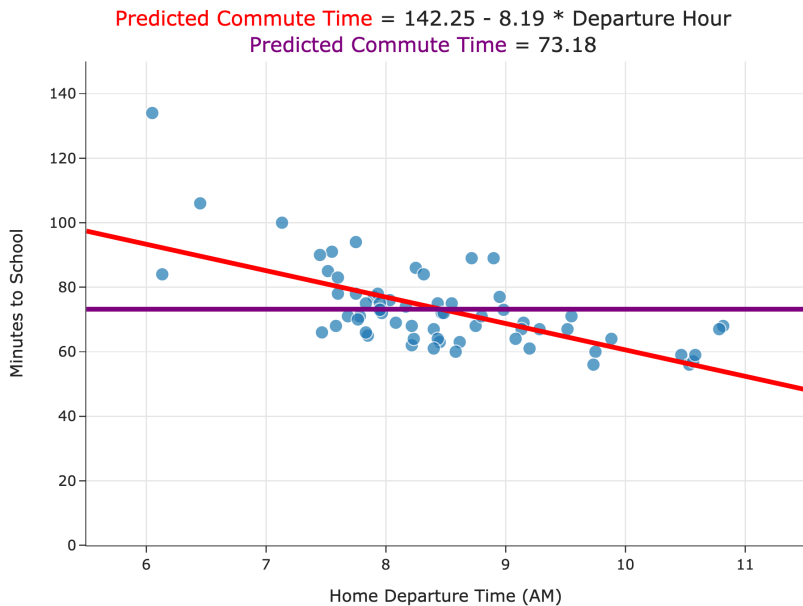
$$w_1^* = \frac{\sum_{i=1}^n x_i y_i}{\sum_{i=1}^n x_i^2}$$

- If our model is $H(x) = w_0$, it is a **line that is forced to have a slope of 0, i.e. a horizontal line.** This is the same as the constant model from before.

$$w_0^* = \text{Mean}(y_1, y_2, \dots, y_n)$$

- **Key idea:** w_0^* above is **not** necessarily equal to w_0^* for the simple linear regression model!

Comparing mean squared errors



$$\text{MSE} = \frac{1}{n} \sum_{i=1}^n (y_i - H(x_i))^2$$

- The MSE of the best simple linear regression model is ≈ 97 .
- The MSE of the best constant model is ≈ 167 .
- The simple linear regression model is a more flexible version of the constant model.

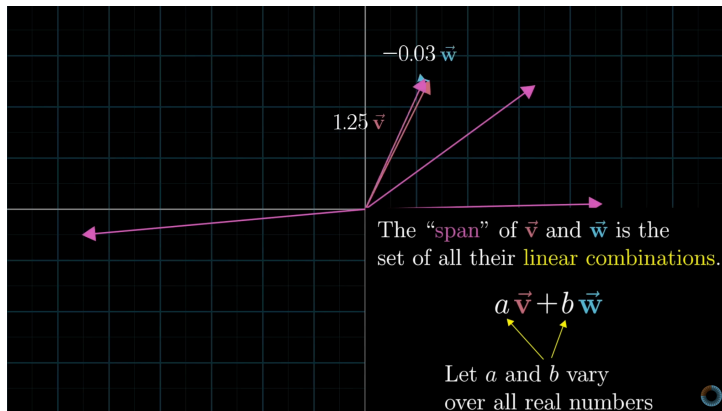
Dot products

Wait... why do we need linear algebra?

- Soon, we'll want to make predictions using more than one feature.
 - Example: Predicting commute times using departure hour and temperature.
- Thinking about linear regression in terms of **matrices and vectors** will allow us to find hypothesis functions that:
 - Use multiple features (input variables).
 - Are non-linear in the features, e.g. $H(x) = w_0 + w_1x + w_2x^2$.
- Before we dive in, let's review.

Spans of vectors

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Warning

- We're **not** going to cover every single detail from your linear algebra course.
- There will be facts that you're expected to remember that we won't explicitly say.
 - For example, if A and B are two matrices, then $AB \neq BA$.
 - This is the kind of fact that we will only mention explicitly if it's directly relevant to what we're studying.
 - But you still need to know it, and it may come up in homework questions.
- We **will** review the topics that you really need to know well.

Vectors

Real #s $\mathbb{R}$ $\mathbb{R}^n$ in later $\rightarrow ||$ n dimensions, n #s in the vector

- A vector in $\mathbb{R}^n$ is an **ordered collection of n numbers.** $\uparrow$ order matters
- We use lower-case letters with an arrow on top to represent vectors, and we usually write vectors as columns.

arrow $\vec{v} = \begin{bmatrix} 8 \\ 3 \\ -2 \\ 5 \end{bmatrix}$ $\neq \begin{bmatrix} 3 \\ 8 \\ 5 \\ -2 \end{bmatrix}$

lowercase $\vec{v}$

row Transpose $\leftarrow$

Another way of writing the above vector is $\vec{v} = [8, 3, -2, 5]^T$.

- Since $\vec{v}$ has four **components**, we say $\vec{v} \in \mathbb{R}^4$.
"elements" (vector) v is in " $\mathbb{R}$ " 4

The geometric interpretation of a vector

- A vector $\vec{v} = \begin{bmatrix} v_1 \\ v_2 \\ \vdots \\ v_n \end{bmatrix}$ is an arrow to the point

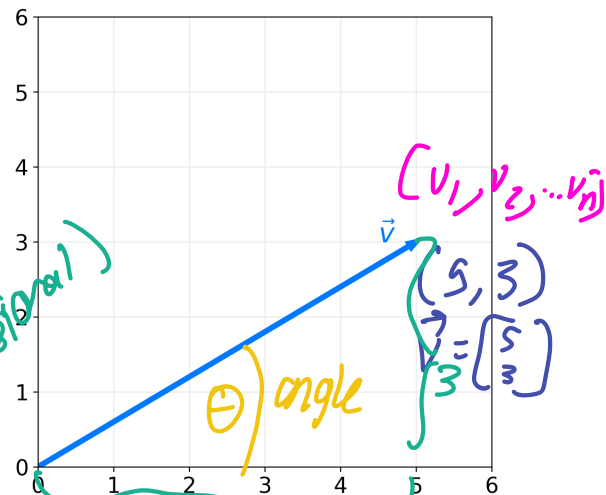
$(v_1, v_2, \dots, v_n)$ from the origin.

- The **length**, or L_2 norm, of $\vec{v}$ is:

$$\|\vec{v}\| = \sqrt{v_1^2 + v_2^2 + \dots + v_n^2}$$

- A vector is sometimes described as an object with a **magnitude/length** and **direction**.

$$\|\vec{v}\| = \sqrt{5^2 + 3^2} = \sqrt{34} \text{ units in same direction}$$



L_3 norm $\sqrt[3]{v_1^3 + \dots}$
 Pythagorean theorem (multidimensional)

θ angle

$$\tan \theta = \frac{3}{5}$$

$$\theta = \tan^{-1}\left(\frac{3}{5}\right)$$

math 18

Dot product: coordinate definition

both have same # of components

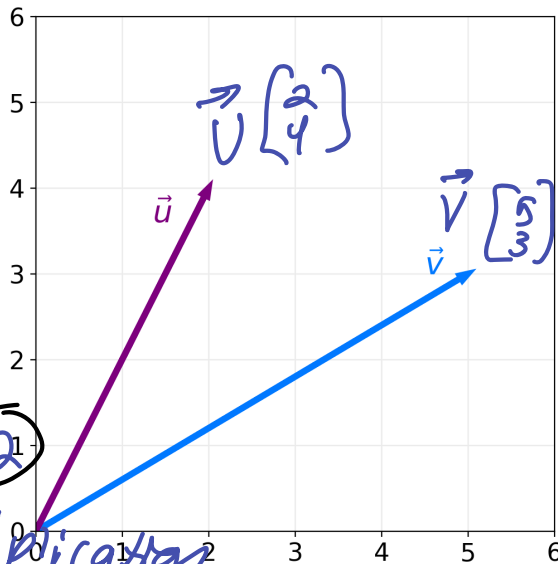
- The **dot product** of two vectors $\vec{u}$ and $\vec{v}$ in $\mathbb{R}^n$ is written as:

$$\vec{u} \cdot \vec{v} = \vec{u}^T \vec{v}$$

- The computational definition of the dot product:

$$\vec{u} \cdot \vec{v} = \sum_{i=1}^n \underline{u_i v_i} = u_1 v_1 + u_2 v_2 + \dots + u_n v_n$$

- The result is a **scalar**, i.e. a single number.



$$\vec{u} \cdot \vec{v} = (2)(5) + (4)(3) = 10 + 12 = 22$$

no dot

$$\vec{u}^T \vec{v} = \begin{bmatrix} 2 & 4 \end{bmatrix} \begin{bmatrix} 5 \\ 3 \end{bmatrix}$$

$1 \times 2 \rightarrow 2 \times 1$

vector multiplication

$$= \begin{bmatrix} 22 \end{bmatrix}$$

1×1

Scalar = just #

Question 🤔

Answer in chat.

Which of these is another expression for the length of $\vec{v}$?

• ~~A. $\vec{v} \cdot \vec{v}$~~

• ~~B. $\sqrt{\vec{v}^2}$~~

• C. $\sqrt{\vec{v} \cdot \vec{v}}$

• ~~D. $\vec{v}^2$~~

• ~~E. More than one of the above.~~

$$\|\vec{v}\| = \sqrt{v_1^2 + \dots + v_n^2} \quad \vec{v} = \begin{bmatrix} v_1 \\ v_2 \\ \vdots \\ v_n \end{bmatrix}$$

$$\sqrt{v^2} \neq \sqrt{\vec{v} \cdot \vec{v}}$$

$$v^2 = \vec{v} \cdot \vec{v} \quad (2 \times 1 * 2 \times 1)$$

$$\begin{aligned} \vec{v} \cdot \vec{v} &= v_1 v_1 + \dots + v_n v_n \\ &= v_1^2 + \dots + v_n^2 \end{aligned}$$

$$\begin{bmatrix} a & b \end{bmatrix} \begin{bmatrix} a \\ b \end{bmatrix}$$

$$\vec{v} \vec{v} \rightarrow 2 \times 1 \leftarrow 2 \times 1 \text{ undefined}$$

$$\vec{v} \cdot \vec{v} = \vec{v}^T \vec{v}$$

$$(1 \times 2) \rightarrow (2 \times 1)$$

$$\begin{bmatrix} a & b \end{bmatrix} \begin{bmatrix} a \\ b \end{bmatrix}$$

Dot product: geometric definition

- The computational definition of the dot product:

$$\vec{u} \cdot \vec{v} = \sum_{i=1}^n u_i v_i = u_1 v_1 + u_2 v_2 + \dots + u_n v_n$$

- The geometric definition of the dot product:

$$\vec{u} \cdot \vec{v} = \|\vec{u}\| \|\vec{v}\| \cos \theta$$

where θ is the angle between $\vec{u}$ and $\vec{v}$.

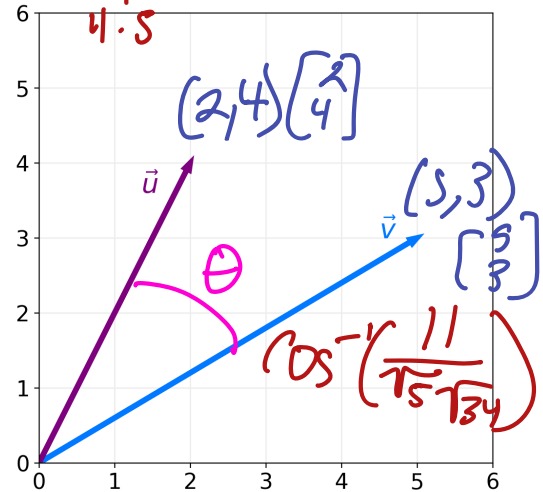
- The two definitions are equivalent! This equivalence allows us to find the angle θ between two vectors.

$$\vec{u} \cdot \vec{v} = (2)(5) + (4)(3) = 22$$

$$\vec{u} \cdot \vec{v} = \|\vec{u}\| \|\vec{v}\| \cos \theta = \sqrt{2^2 + 4^2} \sqrt{5^2 + 3^2} \cos \theta = \sqrt{20} \sqrt{34} \cos \theta$$

$$22 = \sqrt{20} \sqrt{34} \cos \theta$$

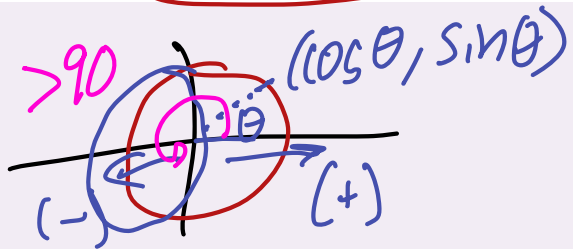
$$\cos \theta = \frac{22}{\sqrt{20} \sqrt{34}} = \frac{22}{2 \cdot \sqrt{5} \sqrt{34}} = \frac{11}{\sqrt{5} \sqrt{34}}$$



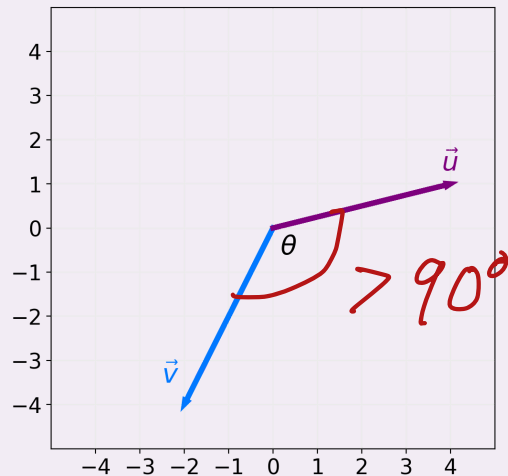
Question 🤔

Answer in chat.

What is the value of θ in the plot to the right?



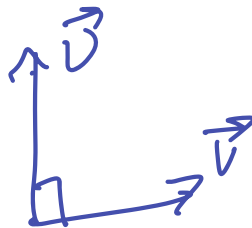
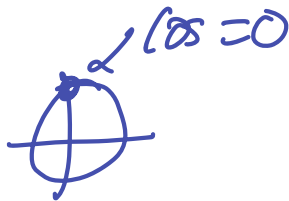
dot product is negative



Vocab

Orthogonal vectors

- Recall: $\cos 90^\circ = 0$.



- Since $\vec{u} \cdot \vec{v} = \|\vec{u}\| \|\vec{v}\| \cos \theta$, if the angle between two vectors is 90° , their dot product is $\|\vec{u}\| \|\vec{v}\| \cos 90^\circ = 0$.

- If the angle between two vectors is 90° , we say they are perpendicular, or more generally, **orthogonal**.

- Key idea:

→ for higher dimensions
word we use instead of $\mathbb{R}^{n>2}$

$$\text{two vectors are orthogonal} \iff \vec{u} \cdot \vec{v} = 0$$

iff = if and only if
"bidirectional statement"

Exercise

Find a non-zero vector in $\mathbb{R}^3$ orthogonal to:

How many of $\vec{v}$ are there?
infinitely many

$$\vec{v} = \begin{bmatrix} 2 \\ 5 \\ -8 \end{bmatrix}$$

easy sol

set 1 elem to 0

$$= \begin{bmatrix} 0 \\ 8 \\ 5 \end{bmatrix} = (0)(2)(8)(5)(5)(8) \\ = 0 + 40 - 40 \\ = 0$$

find solns to

$$2v_1 + 5v_2 - 8v_3 = 0$$

Spans and projections

dot product $\rightarrow$ scalar value

Adding and scaling vectors

- The sum of two vectors $\vec{u}$ and $\vec{v}$ in $\mathbb{R}^n$ is the **element-wise sum** of their components:

$$\begin{bmatrix} u_1 \\ \vdots \\ u_n \end{bmatrix} + \begin{bmatrix} v_1 \\ \vdots \\ v_n \end{bmatrix} = \begin{bmatrix} u_1 + v_1 \\ u_2 + v_2 \\ \vdots \\ u_n + v_n \end{bmatrix}$$

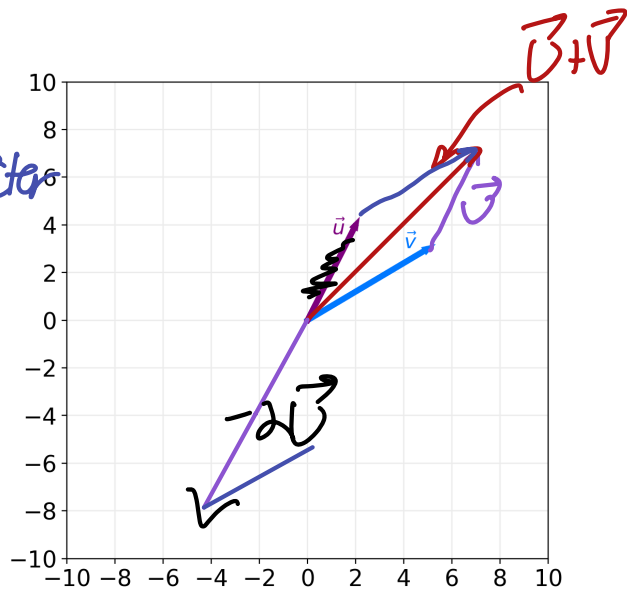
Sum vectors $\rightarrow$ vector

$$\vec{v} = \begin{bmatrix} 2 \\ 4 \end{bmatrix}$$

- If c is a scalar, then:

$$c\vec{v} = \begin{bmatrix} cv_1 \\ cv_2 \\ \vdots \\ cv_n \end{bmatrix}$$

$$-2\vec{v} = \begin{bmatrix} -4 \\ -8 \end{bmatrix}$$



connect tip to tail

Linear combinations

d vectors
n components

$$\begin{bmatrix} v_{1,1} \\ v_{1,2} \\ \vdots \\ v_{1,n} \end{bmatrix}$$

- Let $\vec{v}_1, \vec{v}_2, \dots, \vec{v}_d$ all be vectors in $\mathbb{R}^n$.
- A **linear combination** of $\vec{v}_1, \vec{v}_2, \dots, \vec{v}_d$ is any vector of the form:

$$a_1 \vec{v}_1 + a_2 \vec{v}_2 + \dots + a_d \vec{v}_d$$

where $a_1, a_2, \dots, a_d$ are all scalars.

$$\vec{v}_1 = \begin{bmatrix} 2 \\ 3 \end{bmatrix} \quad \vec{v}_2 = \begin{bmatrix} -1 \\ 4 \end{bmatrix} \quad \vec{v}_3 = \begin{bmatrix} 0 \\ 9 \end{bmatrix}$$

of linear combinations of $\vec{v}_1, \vec{v}_2, \vec{v}_3$

is ∞ many b/c it will always a $\mathbb{R}^2$ vector

$$2\vec{v}_1 - \vec{v}_2 + \frac{1}{9}\vec{v}_3 = \begin{bmatrix} z_1 \\ z_2 \end{bmatrix}$$

add & scalar
multiply

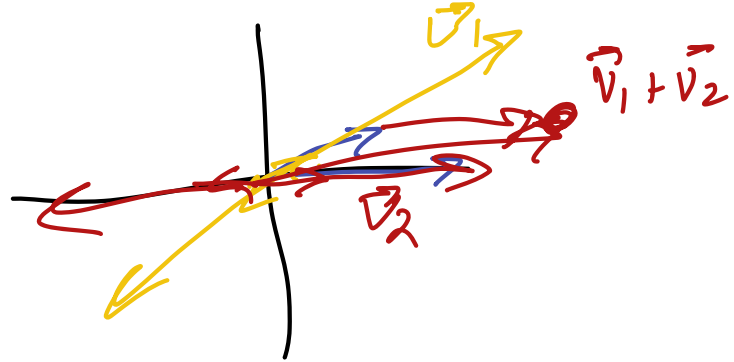
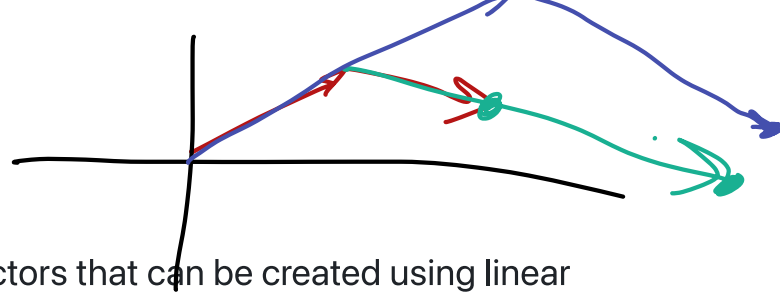
Span

- Let $\vec{v}_1, \vec{v}_2, \dots, \vec{v}_d$ all be vectors in $\mathbb{R}^n$.
- The **span** of $\vec{v}_1, \vec{v}_2, \dots, \vec{v}_d$ is the set of all vectors that can be created using linear combinations of those vectors.
- Formal definition:

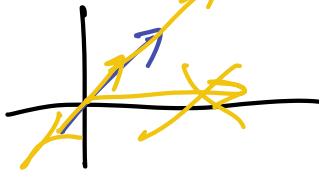
$$\text{span}(\vec{v}_1, \vec{v}_2, \dots, \vec{v}_d) = \{a_1\vec{v}_1 + a_2\vec{v}_2 + \dots + a_d\vec{v}_d : a_1, a_2, \dots, a_n \in \mathbb{R}\}$$

$$\vec{v}_1 = \begin{bmatrix} 2 \\ 1 \end{bmatrix} \quad \vec{v}_2 = \begin{bmatrix} 5 \\ 0 \end{bmatrix}$$

$\vec{v}_1 + \vec{v}_2$ Span $\mathbb{R}^2$
(watch 3 blue 1 brown)



Exercise



Let $\vec{v}_1 = \begin{bmatrix} 2 \\ -3 \end{bmatrix}$ and let $\vec{v}_2 = \begin{bmatrix} -1 \\ 4 \end{bmatrix}$. Is $\vec{y} = \begin{bmatrix} 9 \\ 1 \end{bmatrix}$ in $\text{span}(\vec{v}_1, \vec{v}_2)$?

If so, write $\vec{y}$ as a linear combination of $\vec{v}_1$ and $\vec{v}_2$.

$$w_1 \vec{v}_1 + w_2 \vec{v}_2 = \begin{bmatrix} 9 \\ 1 \end{bmatrix}$$

plug & chug

When is the span
of 2 $\mathbb{R}^2$ vectors not
all of $\mathbb{R}^2$?

if they are in the same
direction

Projecting onto a single vector

- Let $\vec{x}$ and $\vec{y}$ be two vectors in $\mathbb{R}^n$.

- The span of $\vec{x}$ is the set of all vectors of the form:

$w\vec{x}$

span of a singular vector is that vector times any real #

where $w \in \mathbb{R}$ is a scalar.

- Question:** What vector in $\text{span}(\vec{x})$ is closest to $\vec{y}$?
- The vector in $\text{span}(\vec{x})$ that is closest to $\vec{y}$ is the **projection of $\vec{y}$ onto $\text{span}(\vec{x})$** .

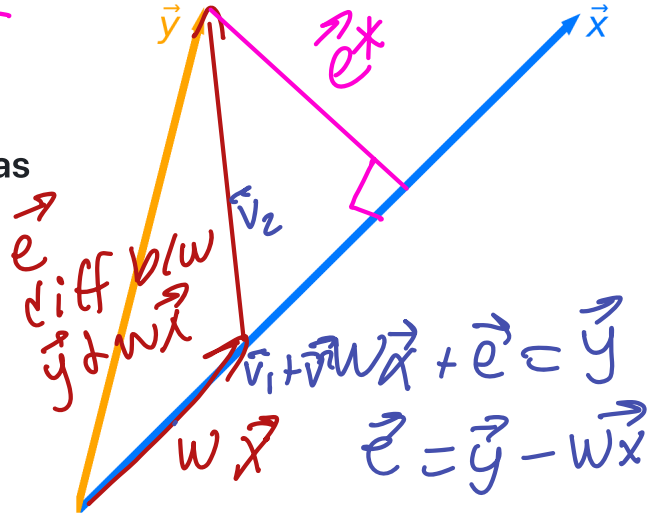


Projection error

vector $\vec{e}$
span of $\vec{x}$

- Let $\vec{e} = \vec{y} - w\vec{x}$ be the projection error:
that is, the vector that connects $\vec{y}$ to $\text{span}(\vec{x})$.
- Goal: Find the w that makes $\vec{e}$ as short as possible.
multiplier
 - That is, minimize:
 $\|\vec{e}\|$
 - Equivalently, minimize:
 $\|\vec{y} - w\vec{x}\|$
- Idea: To make $\vec{e}$ as short as possible, it should be **orthogonal** to $w\vec{x}$.

if $\vec{y}$ is orthogonal to $\vec{x}$ then there's nothing more to do...



Minimizing projection error

- **Goal:** Find the w that makes $\vec{e} = \vec{y} - w\vec{x}$ as short as possible.
- **Idea:** To make $\vec{e}$ as short as possible, it should be **orthogonal to** $w\vec{x}$.
- Can we prove that making $\vec{e}$ orthogonal to $w\vec{x}$ minimizes $\|\vec{e}\|$?

Minimizing projection error

- **Goal:** Find the w that makes $\vec{e} = \vec{y} - w\vec{x}$ as short as possible.
- Now we know that to minimize $\|\vec{e}\|$, $\vec{e}$ must be orthogonal to $w\vec{x}$.
- Given this fact, how can we solve for w ?

Orthogonal projection

- **Question:** What vector in $\text{span}(\vec{x})$ is closest to $\vec{y}$?
- **Answer:** It is the vector $w^*\vec{x}$, where:

$$w^* = \frac{\vec{x} \cdot \vec{y}}{\vec{x} \cdot \vec{x}}$$

- Note that w^* is the solution to a minimization problem, specifically, this one:

$$\text{error}(w) = \|\vec{e}\| = \|\vec{y} - w\vec{x}\|$$

- We call $w^*\vec{x}$ the **orthogonal projection of $\vec{y}$ onto $\text{span}(\vec{x})$** .
 - Think of $w^*\vec{x}$ as the "shadow" of $\vec{y}$.

Exercise

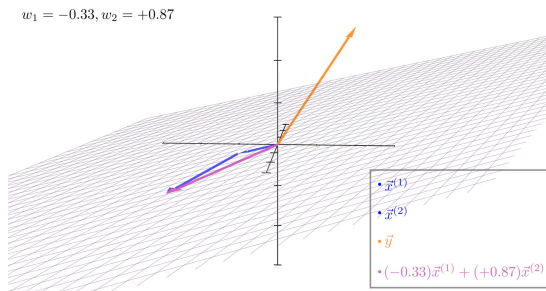
Let $\vec{a} = \begin{bmatrix} 5 \\ 2 \end{bmatrix}$ and $\vec{b} = \begin{bmatrix} -1 \\ 9 \end{bmatrix}$.

What is the orthogonal projection of $\vec{a}$ onto $\text{span}(\vec{b})$?

Your answer should be of the form $w^*\vec{b}$, where w^* is a scalar.

Moving to multiple dimensions

- Let's now consider three vectors, $\vec{y}$, $\vec{x}^{(1)}$, and $\vec{x}^{(2)}$, all in $\mathbb{R}^n$.
- Question:** What vector in $\text{span}(\vec{x}^{(1)}, \vec{x}^{(2)})$ is closest to $\vec{y}$?
 - Vectors in $\text{span}(\vec{x}^{(1)}, \vec{x}^{(2)})$ are of the form $w_1 \vec{x}^{(1)} + w_2 \vec{x}^{(2)}$, where $w_1, w_2 \in \mathbb{R}$ are scalars.
- Before trying to answer, let's watch  [this animation that Jack, one of our tutors, made.](#)



Minimizing projection error in multiple dimensions

- **Question:** What vector in $\text{span}(\vec{x}^{(1)}, \vec{x}^{(2)})$ is closest to $\vec{y}$?

- That is, what vector minimizes $\|\vec{e}\|$, where:

$$\vec{e} = \vec{y} - w_1 \vec{x}^{(1)} - w_2 \vec{x}^{(2)}$$

- **Answer:** It's the vector such that $w_1 \vec{x}^{(1)} + w_2 \vec{x}^{(2)}$ is **orthogonal** to $\vec{e}$.
- **Issue:** Solving for w_1 and w_2 in the following equation is difficult:

$$\left(w_1 \vec{x}^{(1)} + w_2 \vec{x}^{(2)} \right) \cdot \underbrace{\left(\vec{y} - w_1 \vec{x}^{(1)} - w_2 \vec{x}^{(2)} \right)}_{\vec{e}} = 0$$

What's next?

- It's hard for us to solve for w_1 and w_2 in:

$$\left(w_1 \vec{x}^{(1)} + w_2 \vec{x}^{(2)} \right) \cdot \underbrace{\left(\vec{y} - w_1 \vec{x}^{(1)} - w_2 \vec{x}^{(2)} \right)}_{\vec{e}} = 0$$

- Solution:** Combine $\vec{x}^{(1)}$ and $\vec{x}^{(2)}$ into a single **matrix**, X , and express $w_1 \vec{x}^{(1)} + w_2 \vec{x}^{(2)}$ as a **matrix-vector multiplication**, $X\vec{w}$.
- Next time:** Formulate linear regression in terms of matrices and vectors!